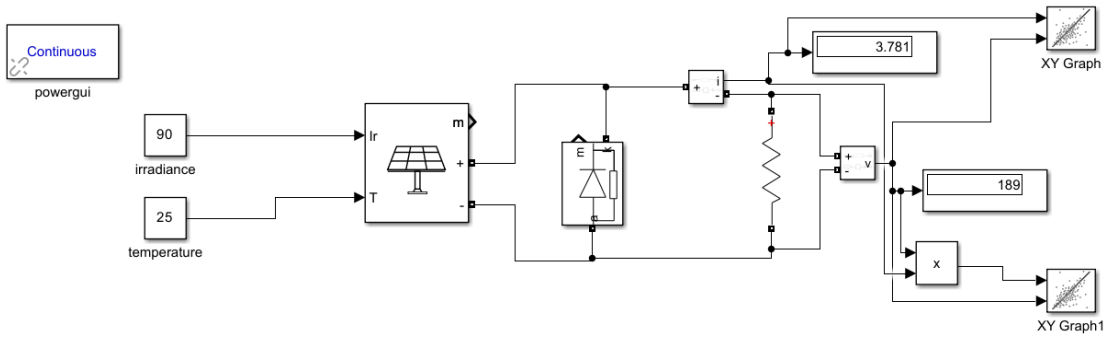
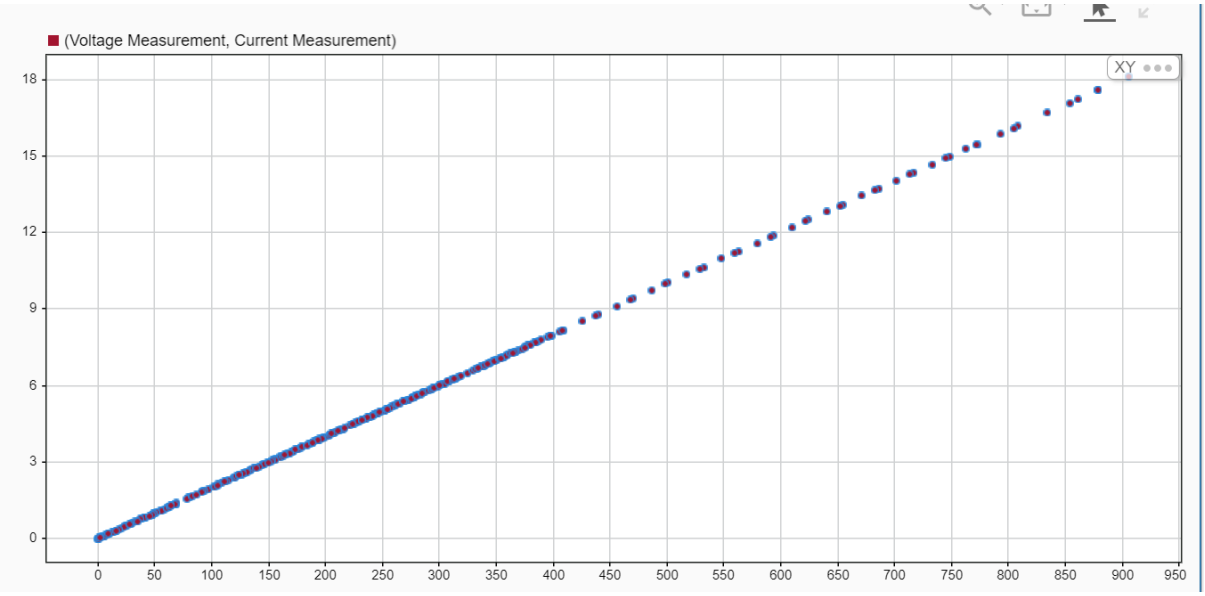
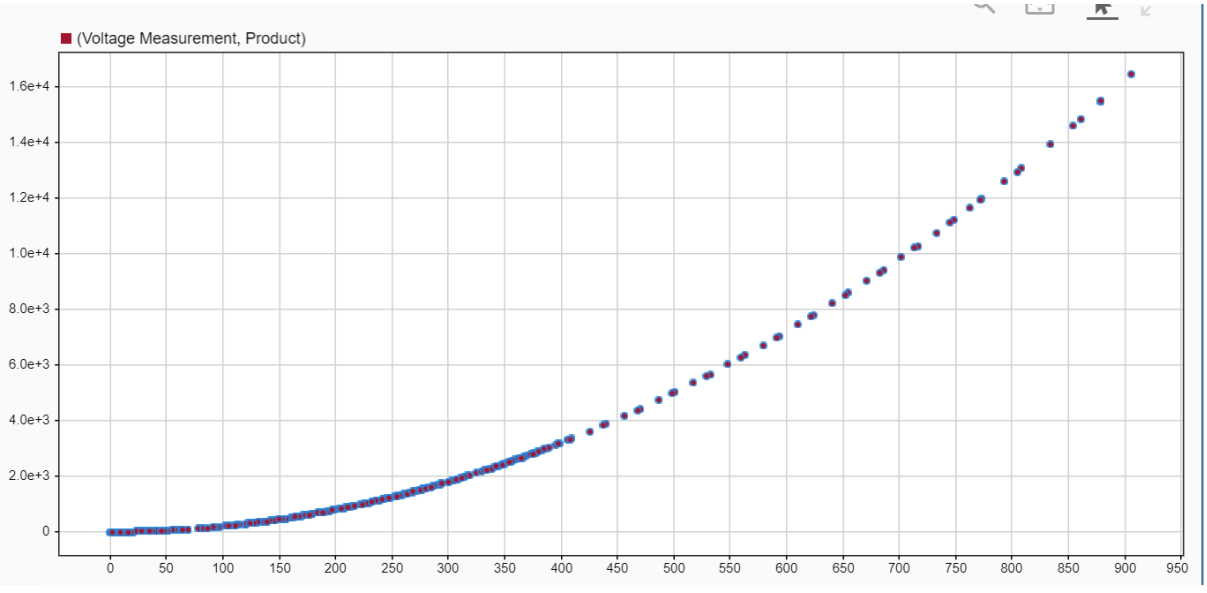
Simulation & Experiment on Characteristics of Solar PV Panel



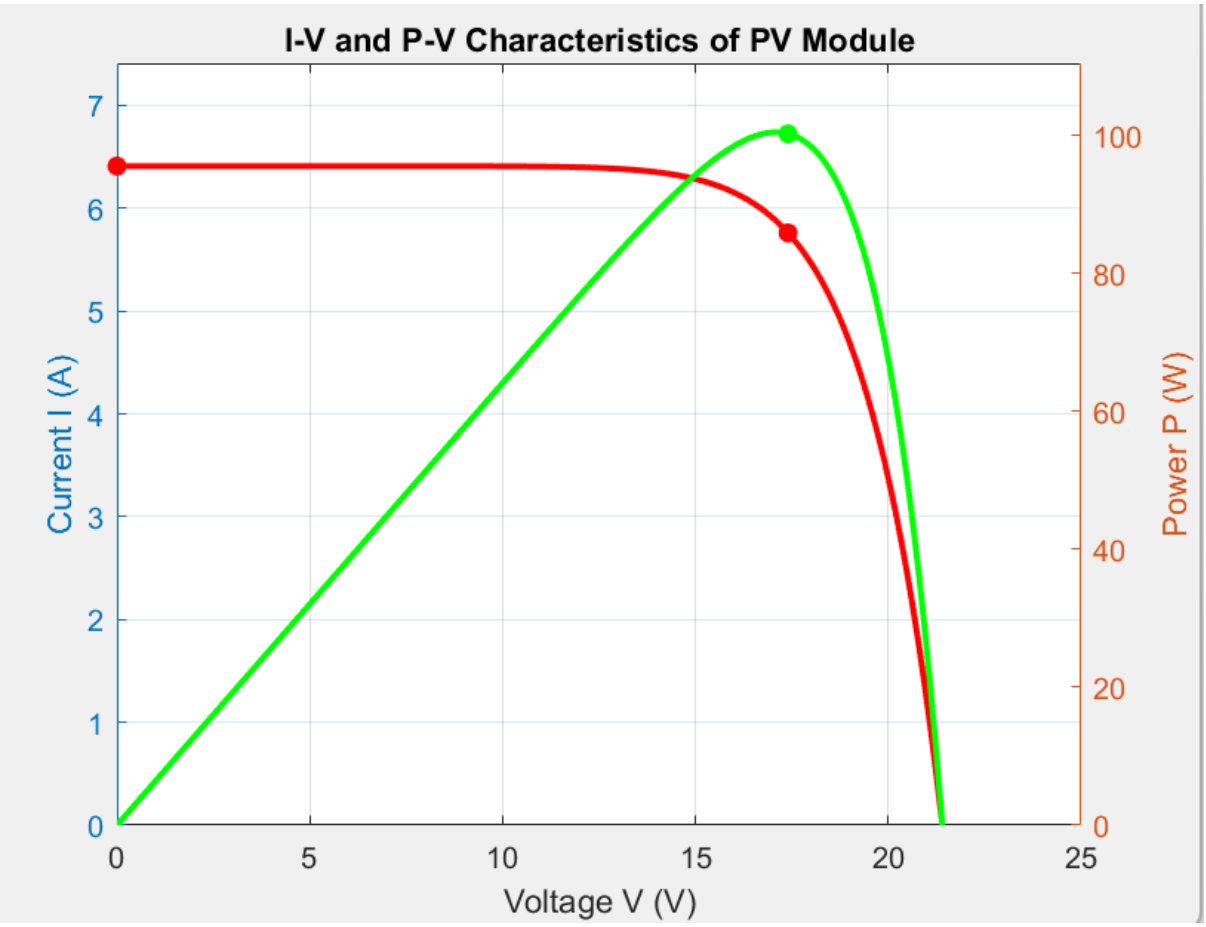
Current Vs Voltage



Power vs Voltage

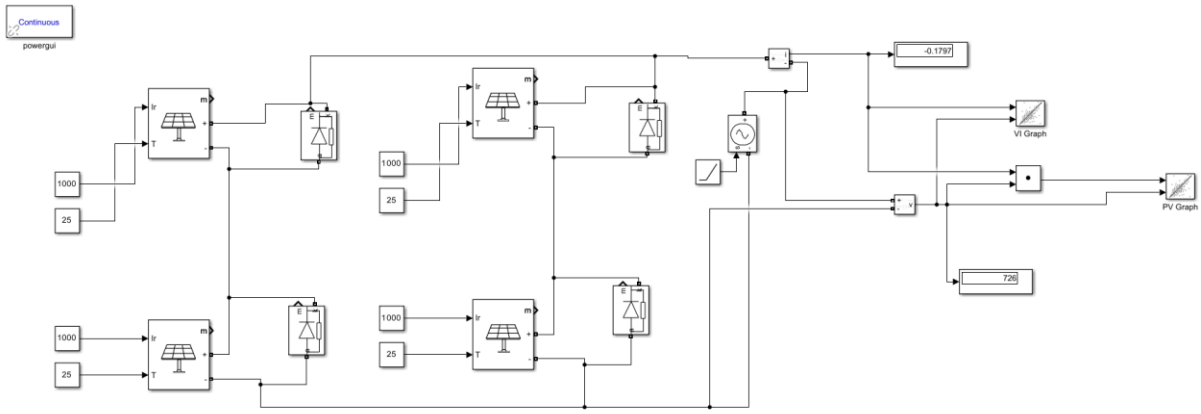


```
clc;
clear;
close all;
Ns = 36;
Np = 1;
Isc = 6.41;
Voc = 21.4;
Imp = 5.76;
Vmp = 17.4;
G = 41;
T = 25;
q = 1.602e-19;
k = 1.38e-23;
a = 1.3;
V = linspace(0,Voc,500);
shape = log(1 - Imp/Isc) / log(Vmp/Voc);
I = Isc * (1 - (V/Voc).^shape);
I(I < 0) = 0;
P = V .* I;
Pmp = Vmp * Imp;
figure;
yyaxis left
plot(V,I,'r','LineWidth',2);
hold on;
plot(0,Isc,'ro','MarkerFaceColor','r');
plot(Vmp,Imp,'ro','MarkerFaceColor','r');
ylabel('Current I (A)');
ylim([0 Isc+1]);
yyaxis right
plot(V,P,'g','LineWidth',2);
hold on;
plot(Vmp,Pmp,'go','MarkerFaceColor','g');
ylabel('Power P (W)');
ylim([0 max(P)+10]);
xlabel('Voltage V (V)');
title('I-V and P-V Characteristics of PV Module');
grid on;
```

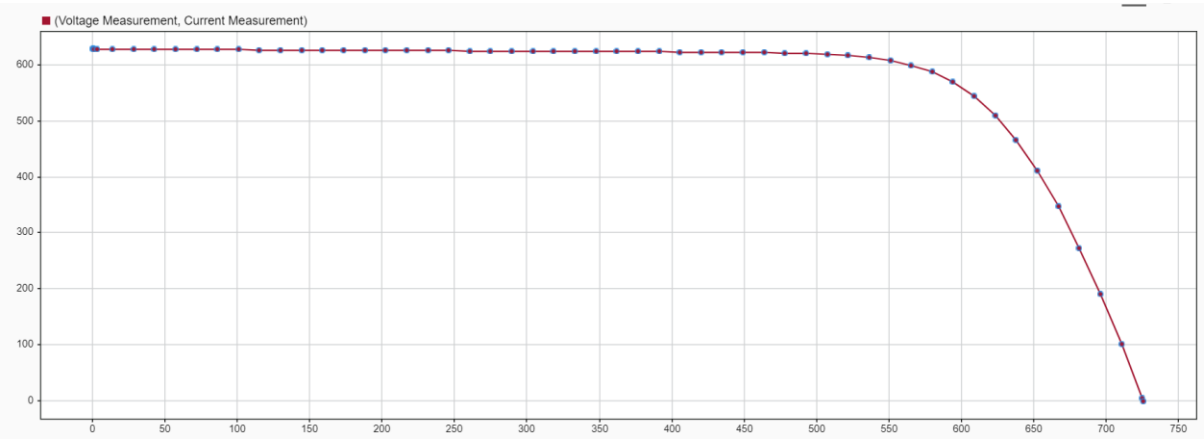

I-V and P-V Characteristics of PV Module
Current I (A)
Power P (W)
Voltage V (V)
0
1
2
3
4
5
6
7
0
20
40
60
80
100
0
5
10
15
20
25

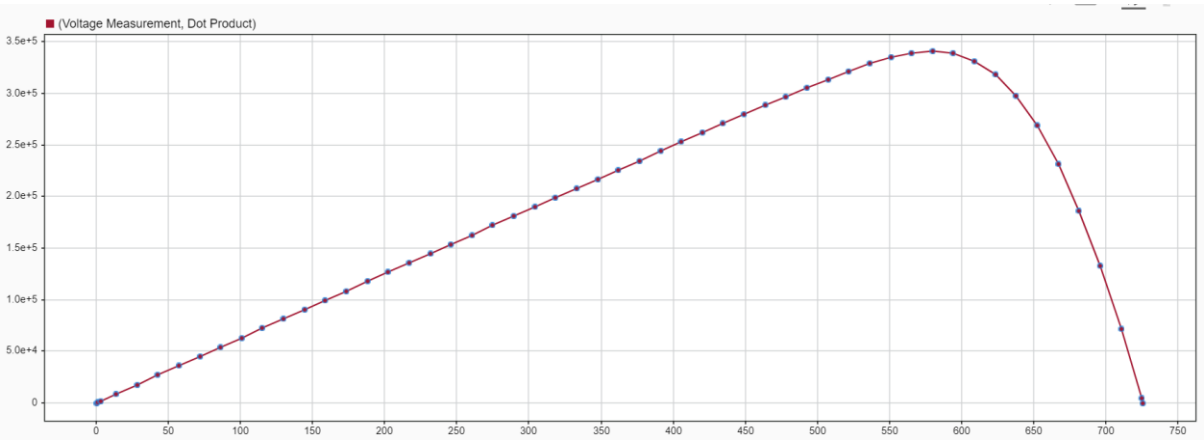
Simulation & Experiment on Characteristics of Shadowing effect in Solar PV Array

Unshaded

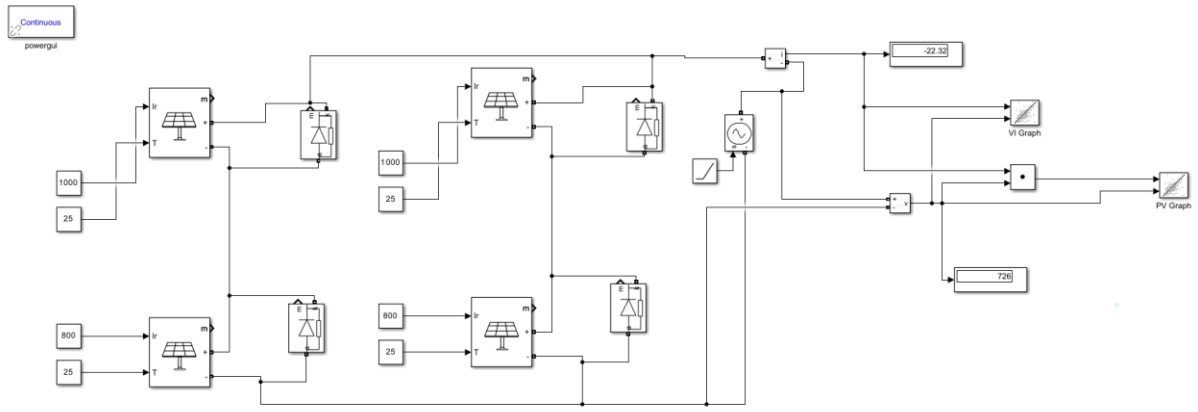


Graphs

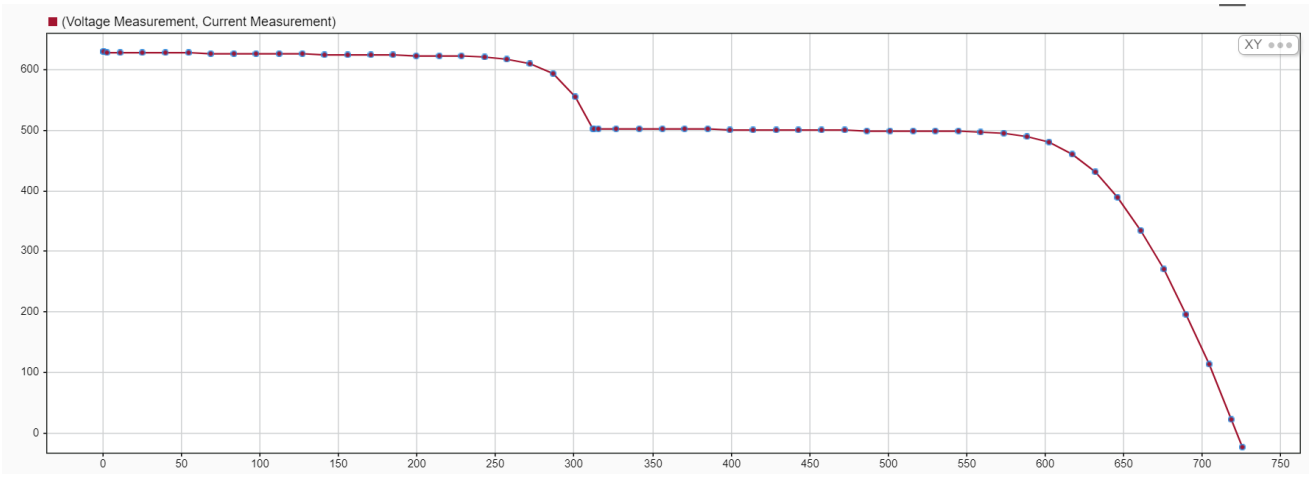


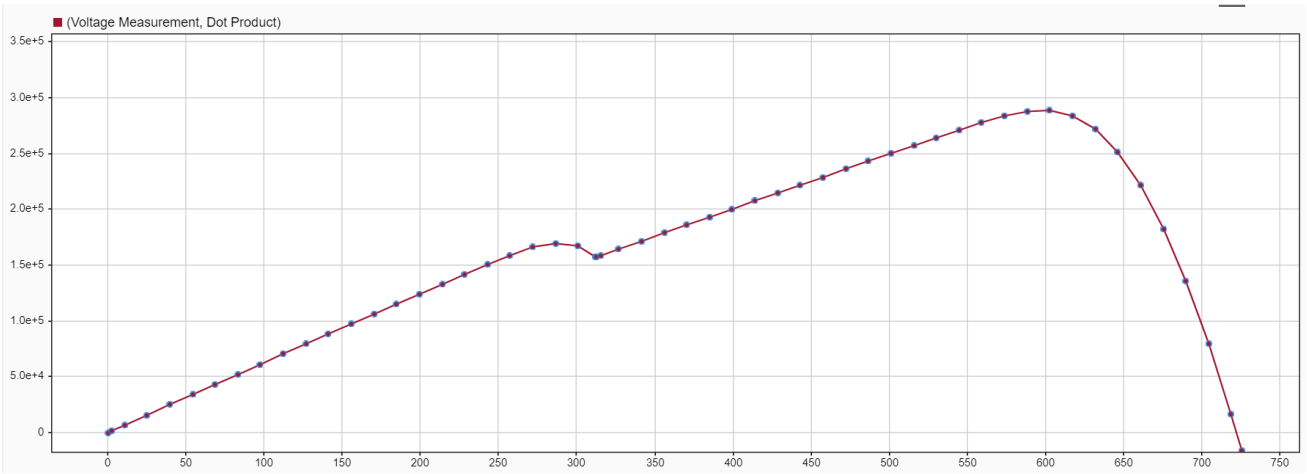


Shaded

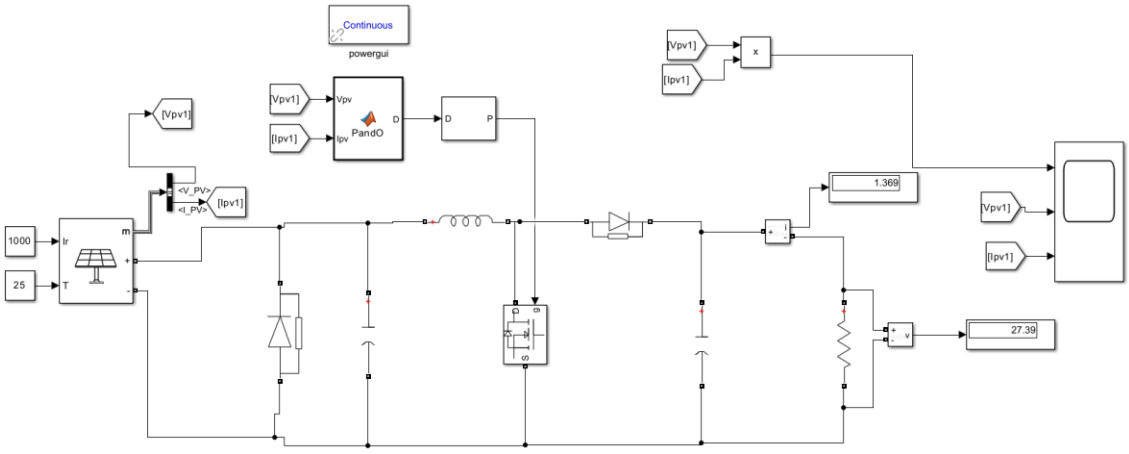


Graphs





Experimental study on Solar PV System with MPPT

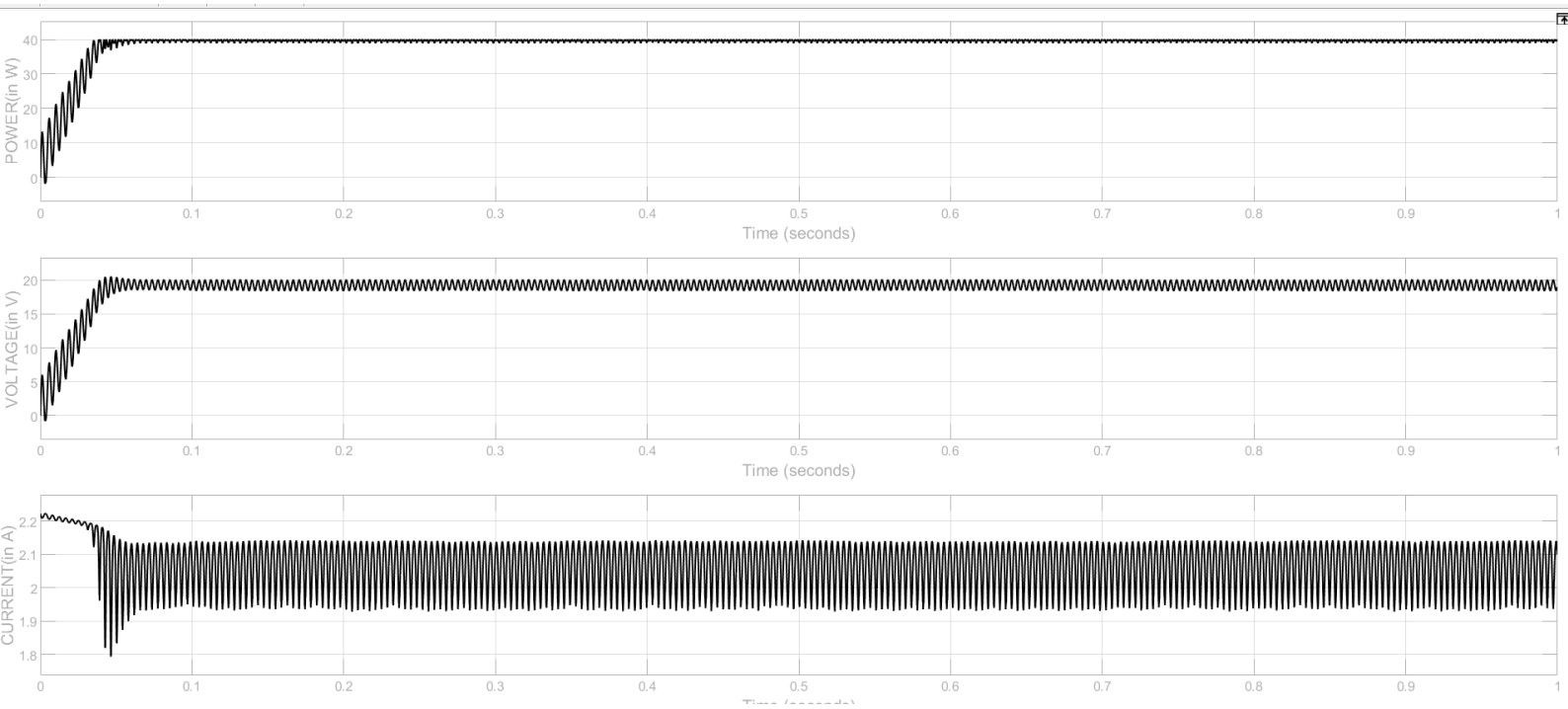


P&O Algorithm

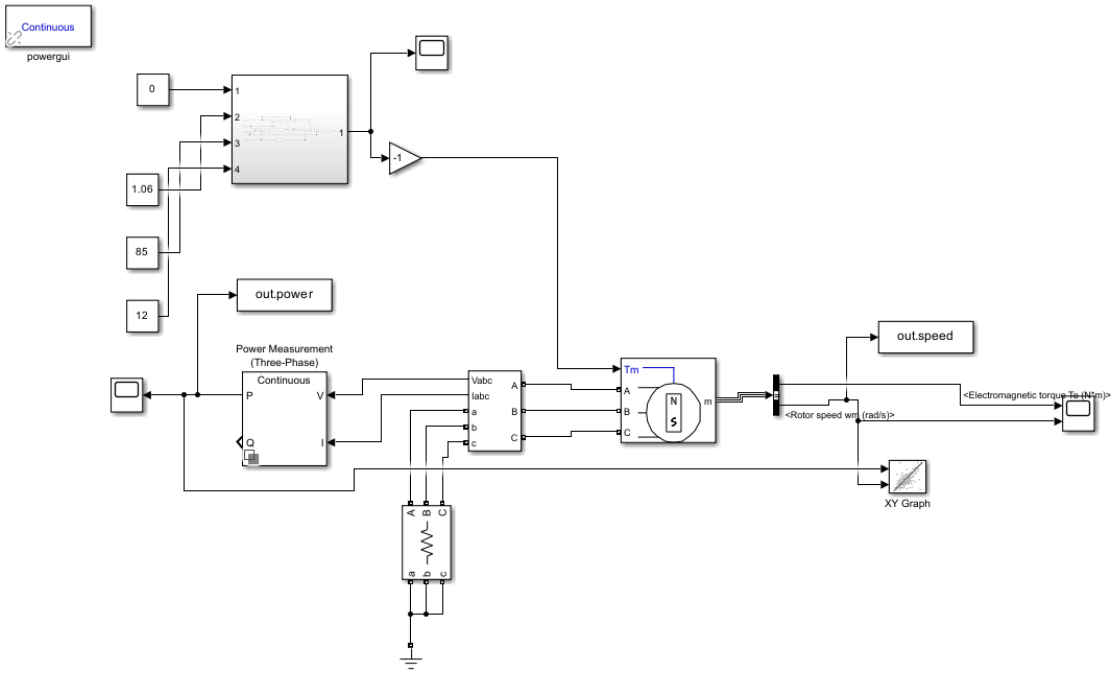
```
function D = PandO(Vpv,Ipv)
persistent Dprev Pprev Vprev
if isempty(Dprev)
    Dprev=0.7;
    Vprev=190;
    Pprev=2000;
end
deltaD=0.001;
Ppv=Vpv*Ipv;
if (Ppv-Pprev)~=0
    if(Ppv-Pprev)>0
        if(Vpv-Vprev)>0
            D=Dprev-deltaD;
        else
            D=Dprev+deltaD;
        end
    else
        if(Vpv-Vprev)>0
            D=Dprev+deltaD;
```

```
        else
            D=Dprev-deltaD;
        end
    end
else
    D=Dprev;
end
Dprev=D;
Vprev=Vpv;
Pprev=Ppv;
```

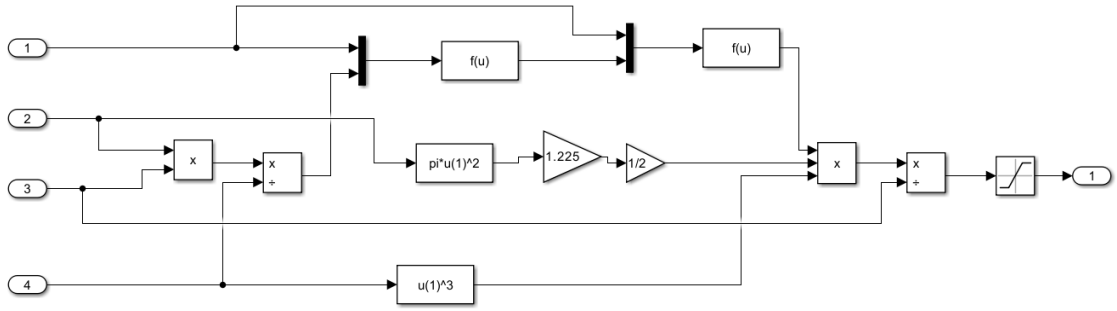
Graph



Simulation & Experimental study on Wind Energy Generator



Wind Turbine

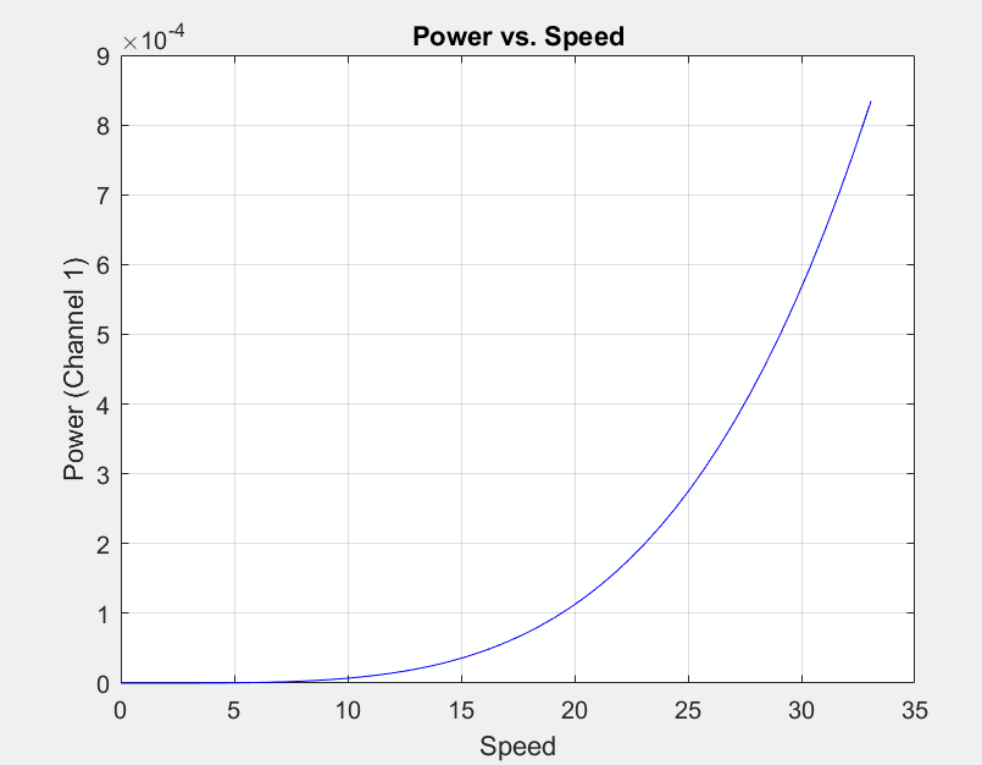


Code

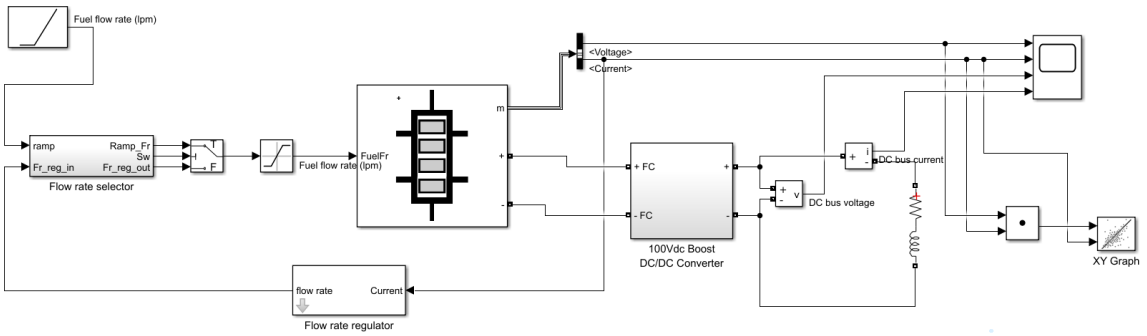
```
X_speed = speed.Data;
Y_matrix = power.Data;
if size(Y_matrix, 1) == 4
    Y_matrix = Y_matrix';
end
Y_power_single = Y_matrix(:, 1);
figure;
plot(X_speed, Y_power_single, 'b');
grid on;
```

```
xlabel('Speed');
ylabel('Power (Channel 1)');
title('Power vs. Speed');
```

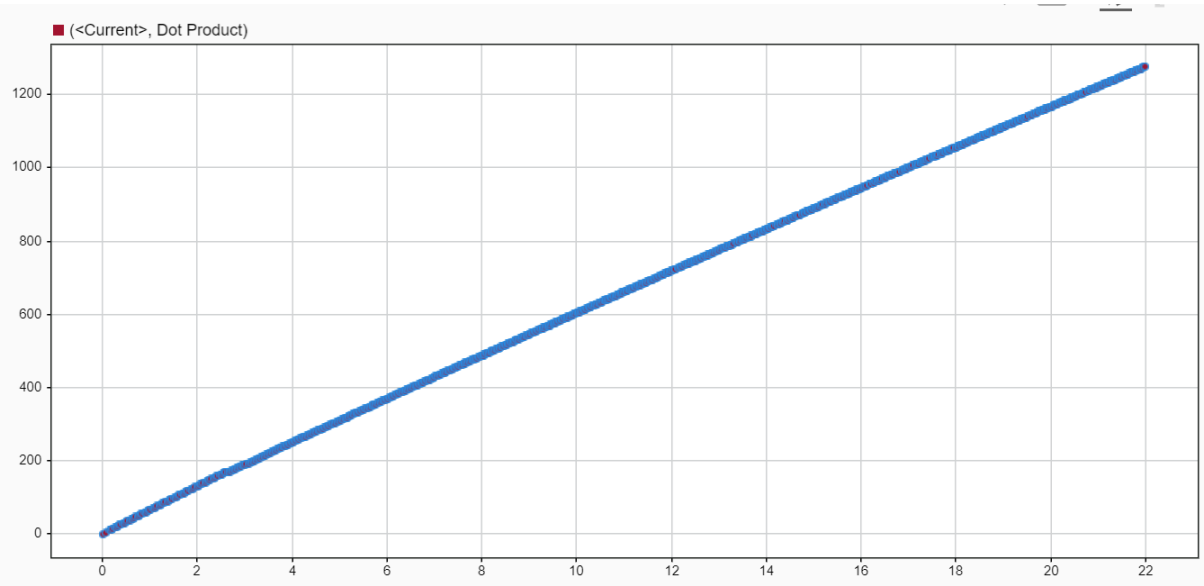
Output Graph



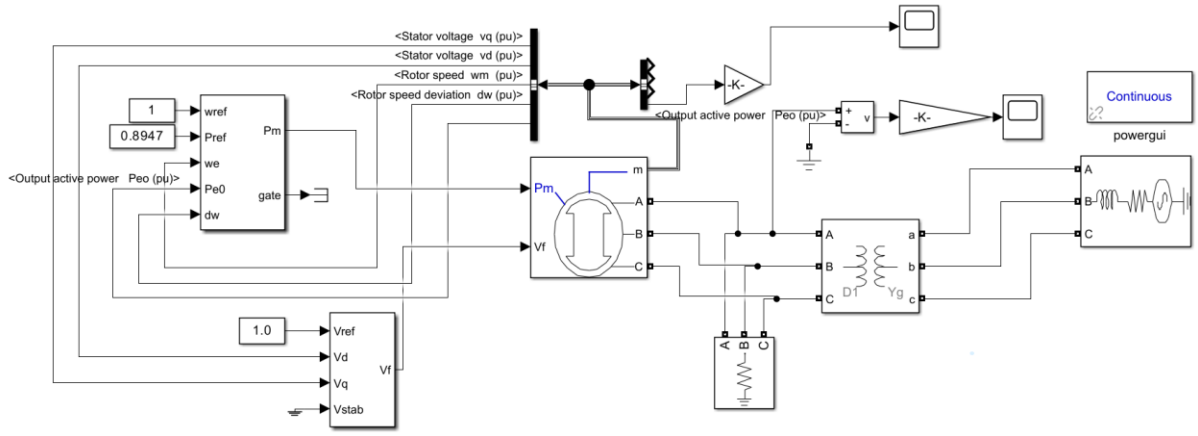
Experiment on Performance Assessment of Fuel Cell



Graph



Simulation study on Hydel Power



Graph

